CLOUD COMPUTING
UNIT III & CLOUD DOCKER

TOPIC: INTRODUCTION TO CLOUD DOCKER

1. INTRODUCTION TO CLOUD DOCKER:

- Docker is a set of Platforms as a service (PaaS) products that use Operating system-level virtualization to deliver software in packages called containers.
- Containers are isolated from one another and bundle their own software, libraries, and configuration files; they can communicate with each other through well-defined channels.
- All containers are run by a single operating system kernel and therefore use fewer resources than a virtual machine.
- Docker is an open-source containerization platform by which you can pack your application and all its dependencies into a standardized unit called a container.
- Containers are light in weight which makes them portable and they are isolated from the underlying infrastructure and from each other container.
- You can run the docker image as a docker container in any machine where docker is installed without depending on the operating system.

1.1 Why Docker is popular?

- Docker gained its popularity due to its impact on the software development and deployment.
- The following are the some of the main reasons for docker becoming popular:

1. **Portability:** Docker facilitates the developers in packaging their applications with all dependencies into a single lightweight containers. It facilities in ensuring the consistent performance across the different computing environments.

2. **Reproducibility:** Through encapsulating the applications with their dependencies within a container it ensures in software setups remaining consistent across the development, testing and production environments.

3. **Efficiency:** Docker through its container based architecture it optimizes the resource utilization. It allows the developers to run the multiple isolated applications on a single host system.

4. **Scalability:** Docker's scalability features facilitated the developers in making easier of their applications handling at time of workloads increment.

1.2 Key Components of Docker

The following are the some of the key components of Docker:

- **Docker Engine:** It is a core part of docker, that handles the creation ana management of containers.
- **Docker Image:** It is a read-only template that is used for creating containers, containing the application code and dependencies.
- **Docker Hub:** It is a cloud based repository that is used for finding and sharing the contaienr images.
- **Dockerfile:** It ia script that containing instructions to build a docker image.
- **Docker Registry**: It is a storage distribution system for docker images, where you can store the images in both public and private modes.

1.3 What is a Dockerfile?

- The Dockerfile uses DSL (Domain Specific Language) and contains instructions for generating a Docker image.
- Dockerfile will define the processes to quickly produce an image. While creating your application, you should create a Dockerfile in order since the **Docker daemon** runs all of the instructions from top to bottom.
- The Docker daemon, often referred to simply as "Docker," is a background service that manages Docker containers on a system.

- It is a text document that contains necessary commands which on execution help assemble a Docker Image.
- Docker image is created using a Dockerfile.

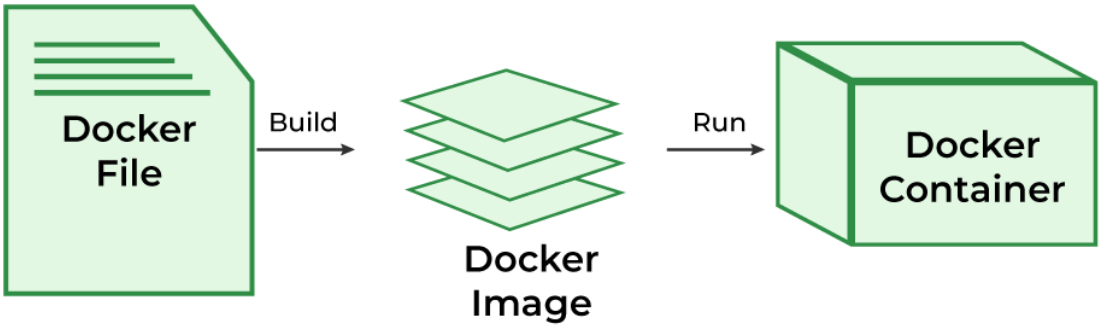

Fig 1.1 Docker File

1.4 Docker Commands

- Through introducing the essential docker commands, docker became a powerful software in streamlining the container management process.
- It helps in ensuring a seamless development and deployment workflows.
- The following are the some of docker commands that are used commonly:

- **Docker Run:** It used for launching the containers from images, with specifying the runtime options and commands.
- **Docker Pull:** It fetches the container images from the container registry like Docker Hub to the local machine.
- **Docker ps**: It helps in displaying the running containers along with their important information like container ID, image used and status.
- **Docker Stop**: It helps in halting the running containers gracefully shutting down the processes within them.
- **Docker Start:** It helps in restarting the stopped containers, resuming their operations from the previous state.

- **Docker Login:** It helps to login in to the docker registry enabling the access to private repositories.

1.5 What is Cloud Docker?

- Cloud Docker refers to the utilization of docker containers within the cloud environments.
- It facilitates the developers in deploying and managing containerized applications effectively across multiple cloud platforms. Cloud Docker facilitates providing encapsulation to the applications and their dependencies.
- It enhances features such as scalability, portability, and efficiency in cloud deployments. It streamlines the development workflows with rapid deployment of applications.

1.6 Terminologies of Docker Cloud

The following are the terminologies of Docker Cloud:

- **Docker ID:** Docker ID provides access to Docker services like Docker Cloud, Docker Hub, Docker storage, and some beta programs.
- **Docker Client:** It's command line instruction.
- **Docker Daemon:** Docker daemon running on the host system, the user can't interact directly with the Daemon.
- **Docker Hub:** It's a clustered resources technique while working with components of docker technology.
- **Docker Compose:** Docker-compose is designed to run multiple containers as a single service. it's written in YAML.
- **Docker Storage:** Docker Store is the best way to distribute and sell your Dockerized content.
- **Docker Server:** Docker daemon is used to interact with the operating system and perform services.

- **Docker Runtime:** The only environmental configuration needed on the server

1.7 Why Use Docker?

The following are the reasons for the usage of Docker:

- **Consistency:** Docker provides the consistency across the platforms from development to the production on reducing the risk of deployment issues due to environment dependencies.
- **Portability:** Docker containers encapsulate the applications and their dependencies making them highly portable across the different systems and environments. It provides the seamless deployments across the various infrastructures.
- **Resource Efficiency:** Docker containers shares the host Operating System's kernel, resulting in the lightweight, efficient resource utilization compared to the traditional virtual machines.

1.8 Why Docker Build Cloud?

The following are the reasons for the usage of Docker Build Cloud:

- **Scalability:** It facilitates easy scaling of the applications in the cloud with rapid horizontal and vertical scaling on accomidating with various workloads.
- **Flexibility**: It provides the flexibility in the deployment of microservices architecture in the cloud with building, deploying and managing the complex distribution systems.
- **Orchestration:** It provides the seamless integration with cloud-native orchestration tools like kubernetes, efficient management of containerizied applications, autoscaling over cluster.

1.9 How Docker Build Cloud Works?

- Docker builds cloud effectively utilize the containerization technology in streamlining the deployment and management of the applications in the cloud.
- The following discuss how it works in detail:

- **Containerization:** It provides the containerization to the applications by packaging all its dependencies into a light weighted software and facilitates with easy portability of containers, ensuring the consistency and comptability.
- **Scalability:** It provides easy scaling of applications in the cloud, with quick adjustment of resources to meet the changing demands without need of manual interventions.
- **Orchestration:** It integrates with cloud-native orchestration tools like Kubernetes which automates the deployment, scaling of applications across clusters of virtual machines or physical servers.
- **DevOps Integration:** It adopts to the practices of DevOps by providing tools for continuous integration and continuous deployment (CI/CD). It supports the teams in automating the pipelines with the release of updates and new features.

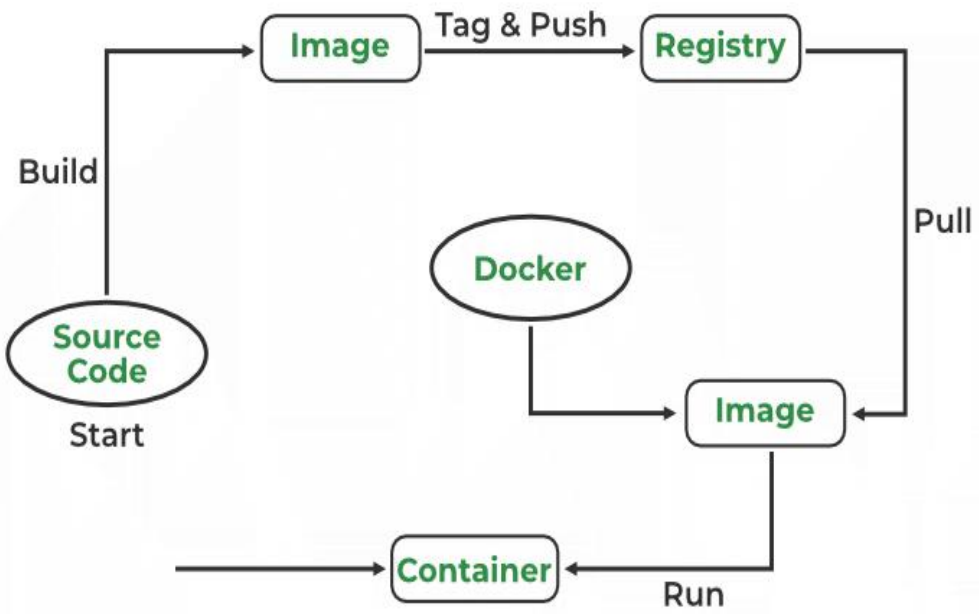

Fig 1.2 Working of Docker Cloud

1.10 Operations of Docker Cloud

The following are the operations of Docker Cloud:

1. Nodes: Connection between the cloud services provider such as MS Azure and AWS to run up containers.
2. Data Processing: Provide big data processing as a service package.
3. Cloud Repository: Provide a platform where we can store a number of versions in a specific Docker Image.
4. Continuous Implementation: Connect with GitHub and build a continuous integration pipeline.
5. App Deployment Scale the Infrastructure and Containers with deployment.
6. Consist Deployment it can be automated deployment.

1.11 Docker Cloud Features

The folowing are the features of Docker Cloud:

- Container Orchestration: It provide the feature such as Container orchestration to orchestrate the containers at scale and facilities the users to deploy and manage scale applications across the multiple cloud environments seamlessly.
- Monitoring and Logging: It offers the monitoring and logging capabilities with the insights of container performance, resource utilization helping users in identifying and resolving the issues proactively.
- Security: It provides the features of securing the containerized applications such as image scanning for vulnerabilities, access controls and network policies.

1.12 Advantages of Docker

- Docker has become popular nowadays because of the benefits provided by Docker containers.
- The main advantages of Docker are:

1. **Speed** – The speed of Docker containers compared to a virtual machine is very fast. The time required to build a container is very fast because they are tiny and lightweight. Development, testing, and deployment can be done faster as containers are small. Containers can be pushed for testing once they have been built and then from there on to the production environment.
2. **Portability** – The applications that are built inside docker containers are extremely portable. These portable applications can easily be moved anywhere as a single element and their performance also remains the same.
3. **Scalability** – Docker has the ability that it can be deployed on several physical servers, data servers, and cloud platforms. It can also be run on every Linux machine. Containers can easily be moved from a cloud environment to a local host and from there back to the cloud again at a fast pace.
4. **Density** – Docker uses the resources that are available more efficiently because it does not use a hypervisor. This is the reason that more containers can be run on a single host as compared to virtual machines. Docker Containers have higher performance because of their high density and no overhead wastage of resources.

DISCUSSION QUESTIONS:

1. Identify the purpose of Docker in Cloud Computing.

2. List the key components of Docker.

3. Interpret the role of Docker File.

4. List the Docker commands and its uses.

5. Illustrate the operations of Cloud Docker.